Digital Material Passport

ID **DMP-METAL-002** - *Version* **1.0.0**
Issue Date **2025-05-14** - *Certificate Type* **EN 10204 3.1**

Manufacturer

ACME Metal Works GmbH
Industrial Park 123
52066 Aachen
DE
quality@acme-metal.example.com

Customer

Engineering Solutions Ltd.
Tech Park Way 45
Cardiff
CF14 5DU
GB
procurement@engisolutions.example.com

Business Transaction

Order	PO-78902 · **2000 kg** Pos. 10 · 2025-04-21
Delivery	DN-56790 · **2000 kg** Pos. 1 · 2025-05-13

Subcustomer

American Heavy Industries Inc.
5000 Industrial Blvd
Building C
Detroit, MI 48201
US
materials@heavyind.example.com

Goods Receiver

Canadian Construction Corp
777 Construction Ave
Toronto, ON M5H 2N2
CA
logistics@canconstruct.example.ca

Product

Product Name	Structural Steel S420N	*Name (EN)*	1.8902
Batch ID	H-10988-01	*Shape*	Plate - Width 2000 mm - Thickness 25 mm - Length 6000 mm
Surface Condition	Hot-rolled	*Product Norm*	EN 10025-3 (2019)
Weight	2000 kg		
Production Date	2025-05-10		
Country of Origin	DE		

Heat Treatment

Process: **Normalizing** - *Lot:* NORM-2024-0823-A12 - *Furnace:* NF-LINE-02 - *Date:* 2024-08-23

Stage	*Temperature*	*Duration*	*Cooling*	*Atmosphere*
Austenitizing	920 C	60 min		Air
Cooling	20 C		Air	

Chemical Analysis

Heat Number **H-10988** - *Melting Process* **EAF+LF+VD** - *Casting Method* **ContinuousCasting** - *Casting Date* **2025-05-09** - *Sample Location* **Ladle**

Symbol	**C**	**Mn**	**Si**	**P**	**S**	**CEV**
Unit	%	%	%	%	%	%
Min						
Max	0.2	1.6	0.5	0.025	0.015	0.44
Actual	0.16	1.48	0.28	0.016	0.01	0.41

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.41%

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Tensile Strength Test temperature: 20°C Specimen: 1/3R, L	Rm	558 / 562 / 560 MPa - Average 560 - Min 558 - Max 562 - Std Dev 2	520 MPa	680 MPa	EN ISO 6892-1	✓
Yield Strength Test temperature: 20°C, 3 specimens tested	ReH	442 / 445 / 448 MPa - Average 445 - Min 442 - Max 448 - Std Dev 3	420 MPa		EN ISO 6892-1	✓
Elongation after fracture Gauge length 5.65#S_0, 3 specimens tested	A	23 / 24 / 25 % - Average 24 - Min 23 - Max 25 - Std Dev 1	19 %		EN ISO 6892-1	✓
Reduction of Area[1]	Z	60 / 62 / 64 % - Average 62 - Min 60 - Max 64 - Std Dev 2	50 %		EN ISO 6892-1	✓
Charpy V-notch Impact Energy[2] Specimen: Top, T-L	KV	56 / 58 / 60 J - Average 58 - Min 56 - Max 60 - Std Dev 2	40 J		EN ISO 148-1	✓
Charpy V-notch Impact Energy[2] Specimen: Bottom, T-L	KV	54 / 57 / 59 J - Average 56.7 - Min 54 - Max 59 - Std Dev 2.5	40 J		EN ISO 148-1	✓
Brinell Hardness Ball: 10mm, Force: 3000kg, 5 indentations	HBW	183 / 185 / 187 / 184 / 186 HBW - Average 185 - Min 183 - Max 187 - Std Dev 1.58	150 HBW	220 HBW	EN ISO 6506-1	✓

Vickers Hardness 10 kg load, 5 indentations	HV	192 / 195 / 198 / 194 / 196 HV10 - Average 195 - Min 192 - Max 198 - Std Dev 2.35	160 HV10	230 HV10	EN ISO 6507-1	✓
Rockwell Hardness	HR	18 HRC		22 HRC	EN ISO 6508-1	✓
Elastic Modulus	E	210 GPa			EN ISO 6892-1	✓
Strain Hardening Exponent	n	0.18			ASTM E646	✓
Plastic Strain Ratio	r	1.2	1		EN ISO 10113	✓
0.2% Proof Strength[1]	Rp0.2	428 / 430 / 432 MPa - Average 430 - Min 428 - Max 432 - Std Dev 2	400 MPa		EN ISO 6892-1	✓

[1] 3 specimens tested [2] Test temperature: -20°C

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14